# Trade Openness and Consumption Channels in Small Open Economies

## Geoeconomic shocks, small open economies, and policy design

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# The Global Landscape Has Become Harder For Small Open Economies

| **Trade fragmentation** | **Energy and shipping shocks** | **Climate shocks** |
|---|---|---|
| policy uncertainty, re-routing, and slower global trade growth | higher fuel costs, transport disruption, and imported inflation | storms, reconstruction, and more volatile import needs |

**Why this paper is important:** in a small open economy, these shocks arrive through prices, trade flows, and household income at the same time.

# Openness to Trade Matters, But How Exactly?

- Which external shocks hit inflation and consumption first?
- Which households feel the shock mainly through current income rather than asset prices?
- What can monetary or fiscal policy realistically do on their own, and how to best design policies?

This is where DSGE work becomes useful for policy: not as a black box, but as a disciplined way to map shocks into channels and channels into choices.

# Trade Openness and Consumption Channels in Small Open Economies

## Research Question

- Monetary policy and external shocks affect SOEs through IS channel and open-economy relative-price and income effects.
- With a high share of constrained households, these open-economy forces can reshape aggregate consumption dynamics because exchange-rate movements reallocate real income across agents and across borders.
- How does trade openness affects the sensitivity of aggregate consumption in an SOE with heterogeneous households?
- Does greater trade openness amplify or attenuate the response of consumption to changes in the domestic/foreign monetary policy and foreign demand?

# In this paper

- Develop a tractable two-agent NK SOE (TANK-SOE) model.
- Allow for asymmetric trade openness between the home economy and the rest of the world:
  - $\alpha = \frac{M}{A_X}$: import openness in domestic absorption ($A = Y - X + M$)
  - $\alpha^\star = \frac{X}{Y}$: foreign demand weight on home goods
  - $\alpha^\star \neq \alpha$: unbalanced trade configuration
- We also allow for import-substitution and export-demand elasticities to differ.
- We characterize analytically the sensitivity of $\hat{c}_t$ to changes in $\hat{r}_t^r$, $\hat{r}_t^{\star r}$, and $\hat{y}_t^\star$:
  - $\Xi_r = \partial \hat{c}_t / \partial \hat{r}_t^r$: Intertemporal substitution (IS) and expenditure-switching (ES) channels.
  - Depending on the trade balance, heterogeneity, and the elasticities of substitution, the ES channel can amplify or attenuate the IS channel.

# Why Break Balanced Trade?

- Standard SOE NK models summarize openness with one home-bias parameter.
- In SOE NK models, balanced trade ties the import and export sides together in steady state.
- That restriction removes the trade balance as a determinant of monetary transmission.

If $\alpha = \alpha^{\star}$, the model cannot ask how deficits versus surpluses shift the response of consumption to the real rate.

# Import and Export Openness in the Data

## WDI

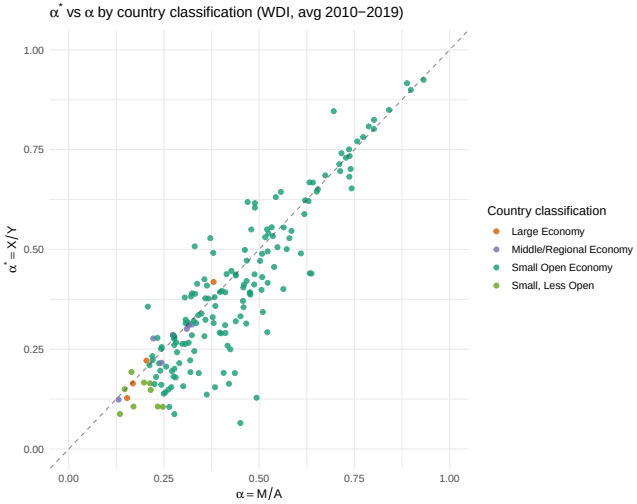
$$A \equiv C + I + G = Y - X + M,$$

$$\alpha \equiv \frac{M}{A},$$

$$\alpha^{\star} \equiv \frac{X}{Y}.$$

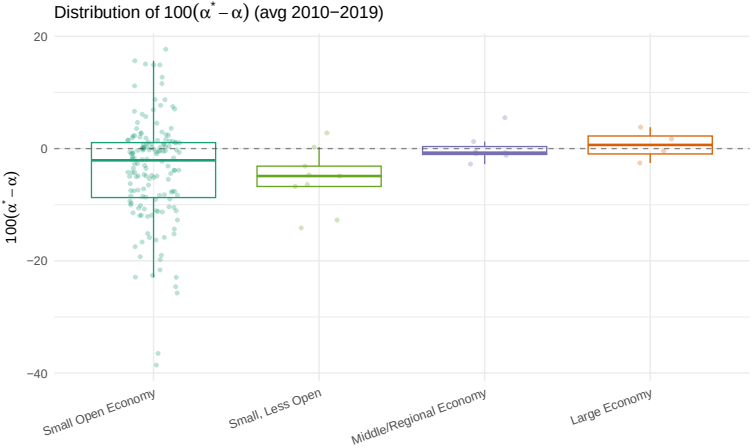
- $\alpha$: absorption-based import openness.
- $\alpha^{\star}$: output-based export openness.
- The gap $\alpha^{\star} - \alpha$ is zero under balanced trade and shifts sign with the trade balance.

# Import and Export Openness: sizeable wedge



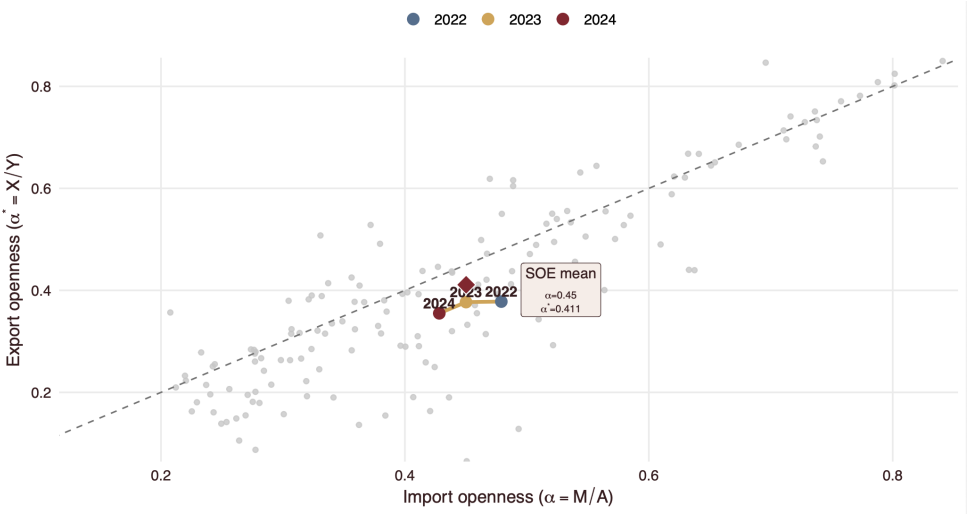

2010–2019 averages of $(\alpha, \alpha^{\star})$ by market class.

# Import and Export Openness: More Dispersion in SOE




Distribution of $100 \times (\alpha^{\star} - \alpha)$. The 45-degree line is the balanced-trade benchmark within the model-consistent region $\alpha^{\star} < \alpha$.

# Import and Export Openness: Jamaica



# Model Environment

- Small open economy New Keynesian model with Calvo price rigidities.
- Two household types:
  - ($U$ - $1-\lambda$) unconstrained households with domestic and foreign bond access;
  - ($C$ - $\lambda$) constrained households who consume current income.
- Distinct openness margins:
  - $\alpha$: import share in the domestic consumption basket;
  - $\alpha^{\star}$: foreign demand weight on home output.
- Distinct trade elasticities $(\eta, \gamma)$ and general IES $\sigma$.
- The rest of the world is a continuum of small open economies where each domestic policy decision does not affect the rest of the world.
- The monetary authority directly controls the real rate.

# Aggregate Euler Equation in the TANK-SOE

$$\hat{c}_t = \mathbb{E}_t \hat{c}_{t+1} - \Xi_{y^\star} \mathbb{E}_t \Delta \hat{y}^\star_{t+1} + \Xi_{r^\star} \hat{r}^{\star r}_t + \Xi_r \hat{r}^r_t - \Xi_{r^\star} \zeta_B B^\star_u \hat{b}^\star_{u,t}$$

with

$$\Xi_r = -\frac{\frac{1-\lambda \frac{1-\alpha}{1-\alpha^\star}}{\sigma} + \frac{\lambda}{1-\alpha^\star} \bar{\psi}}{1-\lambda(1-\alpha)}$$

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$$\Xi_r = -\frac{\frac{1-\lambda \frac{1-\alpha}{1-\alpha^\star}}{\sigma} + \frac{\lambda}{1-\alpha^\star} \bar{\psi}}{1-\lambda(1-\alpha)}$$

- In the RANK limit ($\lambda = 0$) or closed-economy limit ($\alpha = \alpha^\star = 0$), this collapses to the standard $-1/\sigma$ slope.
- All open-economy heterogeneity effects are summarized by the coefficients on $\hat{y}^\star_t$, $\hat{r}^\star_t$, and $\hat{r}_t$.

# The Domestic Real-Rate Coefficient

$$\Xi_r = -\frac{\frac{1-\lambda\frac{1-\alpha}{1-\alpha^\star}}{\sigma} + \frac{\lambda}{1-\alpha^\star}\bar{\psi}}{D}, \qquad D \equiv 1-\lambda(1-\alpha)$$

- $\Xi_r < 0$: a higher domestic real rate lowers aggregate consumption.
- Our focus is on how the magnitude of that slope changes with openness, elasticities, and heterogeneity.
- Separate $\alpha$ and $\alpha^\star$ matter because they enter the numerator and denominator differently.

# Decomposing $\Xi_r$: IS + ES

$$\Xi_r = \underbrace{-\frac{\frac{1-\lambda\frac{1-\alpha}{1-\alpha^\star}}{\sigma}}{D}}_{\Xi_r^{IS}} + \underbrace{-\frac{\lambda\bar{\psi}}{D(1-\alpha^\star)}}_{\Xi_r^{ES}}$$

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**Intertemporal Substitution**

- direct smoothing by unconstrained households;
- scaled down by the income-to-consumption wedge;
- scaled up by the common multiplier $1/D$.

**Expenditure Switching**

- exchange-rate-induced terms-of-trade and demand reallocation;
- operative only when $\lambda > 0$;
- sign governed by $\bar{\psi}$.

# Jamaica Brings The Main External Channels Into One Policy Case

- Domestic spending is import intensive.
- External income is concentrated in tourism and related services.
- Remittances still cushion many households.
- Climate shocks and reconstruction make the external side of policy impossible to ignore.

| | |
|---|---|
| **Import–intensive absorption**<br>M/Y = 48.3% | **Tourism–heavy export base**<br>26.3% of GDP |
| **Large remittance cushion**<br>15.3% of GDP | **Climate exposure is macro–critical**<br>GDP = –4.3% |

# What does this paper imply for Jamaica?

**Spending side**

- How much of expenditure leaks abroad?
- This shapes how much domestic demand feeds back into domestic income.
- It tells us how strong the domestic multiplier can be.

**Income side**

- How much foreign demand becomes domestic income?
- In Jamaica, tourism dominates that channel.
- Remittances add support to household cash flow from outside credit markets.

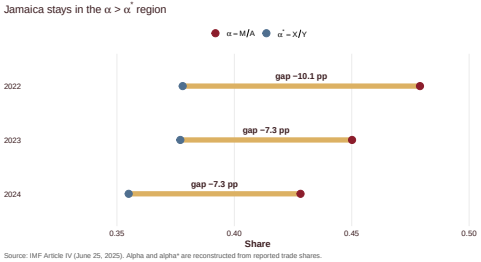
If the spending side is more open than the income side, policy transmission looks different from the closed-economy story many of us teach first.

# Jamaica Is More Open On Import Than On Export

$$A = Y - X + M$$

$$\alpha = \frac{M}{A}, \qquad \alpha^* = \frac{X}{Y}$$

- $\alpha$: import share of expenditure.
- $\alpha^*$: export share of income.
- A one-openness setup effectively collapses the two together.
- Jamaica stays in the $\alpha > \alpha^*$ region through 2022–2024.




Imported costs and FX conditions move consumption before export earnings can fully offset them.

Exact openness mapping

# When $\alpha > \alpha^*$, Monetary Policy Is Not A Closed-Economy Demand Story

| | Closed-economy reflex | Jamaica reading |
|---|---|---|
| Higher real rates | Mainly compress domestic borrowing and spending. | Still compress spending, but part of the response leaks into imports rather than recycling fully into domestic income. |
| Exchange rate | Often a side story. | Immediately affects imported prices, inflation risks, and external competitiveness. |
| Net effect | A fairly clean domestic multiplier. | A smaller domestic multiplier plus larger relative-price and external-income effects. |

That is why the same policy rate can feel weaker for domestic demand but still matter a great deal for inflation and financial conditions.

# The 2026 Jamaica Setting Is A Resilience Test, Not A Normal Cycle

**June 25, 2025**
IMF Article IV: strong buffers, cautious policy, and scope to deepen FX markets over time.

**January 16, 2026**
IMF RFI after Hurricane Melissa: reconstruction, vulnerable households, and urgent balance-of-payments needs.

**March 31, 2026**
BOJ: heightened inflation uncertainty linked to Middle East conflict and energy-related prices.

Reconstruction imports, energy prices, and policy credibility are all part of the same macro problem.

# Post-Disaster Reconstruction

## As Much an Import Shock as a Fiscal Shock

### Current conditions

- Hurricane Melissa shock large enough to trigger the fiscal-rule escape clause and a reconstruction phase.
- Rebuilding raises imports of materials, equipment, fuel, and transport services.
- The World Bank expects the current account to turn negative in 2026 as reconstruction imports rise.

### Why this matters for Jamaica?

- A larger fiscal envelope does not guarantee a large local multiplier when spending leaks abroad quickly.
- Import-heavy recovery can raise inflation and FX pressure even in a weak domestic cycle.
- Measures that use local labor, local logistics, and targeted transfers support domestic income more directly.

The right post-disaster question is spending composition, not just spending size.

# Policy Design Matters

| Policy area | Why it matters in this framework |
|---|---|
| Tourism linkages | More local sourcing raises the share of foreign demand that becomes Jamaican income rather than import leakage. |
| Energy transition | Lower fuel dependence weakens imported-cost pass-through and makes inflation control less hostage to external prices. |
| Targeted household support | When many households are cash-flow sensitive, well-designed transfers stabilize consumption better than blunt subsidies. |
| FX deepening and de-dollarization | Better market depth can improve transmission, but sequencing matters when imported inflation risks remain high. |

In this sense, tourism policy, energy policy, and social protection are not side issues. They help determine how macro stabilization works.

# Jamaica Also Opens A Wider Small-Open-Economy Discussion

- How many small open economies are misread when we summarize openness with one number?
- Which external income channels matter most for policy design: tourism, commodity exports or remittances?
- When shocks are imported, what is the right division of labor between monetary policy, FX management, and structural policy?

Jamaica as a concrete example of a broader question: how do we build policy frameworks for economies that are open in different ways on the spending and income sides?

# Three Takeaways

1. Small open economies are not all open in the same way, and that matters for transmission.
2. Jamaica is a clear case where imported prices, tourism income, remittances, and household heterogeneity all matter together.
3. Good policy design means combining credible monetary & fiscal policy with stronger domestic linkages, lower energy dependence, and better protection for vulnerable households.

Thank you

# Appendix: Jamaica's Openness Wedge Stays Negative In 2022–2024

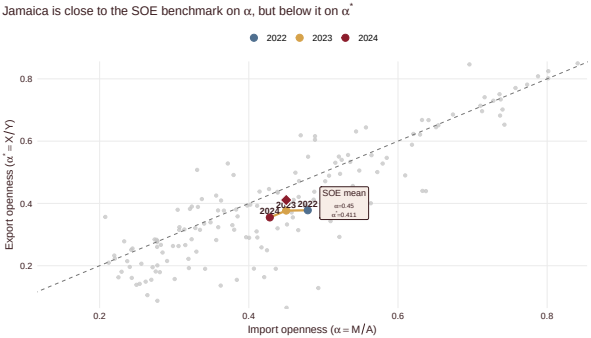
$$A = C + I + G = Y - X + M, \qquad \alpha = \frac{M}{A}, \qquad \alpha^* = \frac{X}{Y}$$

| Year | $X/Y$ | $M/Y$ | $\alpha$ | $\alpha^* - \alpha$ |
|---|---|---|---|---|
| 2022 | 0.378 | 0.572 | 0.479 | -0.101 |
| 2023 | 0.377 | 0.510 | 0.450 | -0.073 |
| 2024 | 0.355 | 0.483 | 0.428 | -0.073 |

- $X/Y$ uses IMF goods and services exports as a share of GDP.
- $M/Y$ uses IMF goods and services imports as a share of GDP.
- Jamaica remains in the trade-deficit configuration highlighted in the paper.

Back to openness wedge

# Appendix: Jamaica On The Broader Small-Open-Economy Map




Sources: WDI–based SOE benchmark from the paper's country summary (2010–2019 averages) and IMF Jamaica data for 2022–2024.

- Jamaica is not unusually closed.
- The more relevant point is that it is more import intensive than its export base would suggest.

Back to discussion slide

# Appendix: Source Anchors Used For The Kingston Talk


- Cantore, Cristiano and Sebastian Diz, *Trade Openness and Consumption Channels in Small Open Economies*.
- Policy memo: *Applying "Trade Openness and Consumption Channels in Small Open Economies" to Jamaica*.
- IMF, *World Economic Outlook*, April 14, 2026.
- WTO, *Trade outlook update*, March 19, 2026.
- IMF, *Executive Board Concludes 2025 Article IV Consultation with Jamaica*, June 25, 2025.
- IMF, *Executive Board Approves a US$415 Million Disbursement to Jamaica to Address the Impact of Hurricane Melissa*, January 16, 2026.
- Bank of Jamaica, *Summary of Decisions – March 2026*, March 31, 2026.

- Repo data anchor: `draft/figures/jamaica_application/jamaica_application_series.csv`.
- Repo comparison anchor: `draft/figures/kappa_gap/kappa_gap_country_summary.csv`.

Back to close